

FSK116: Tutorial Problem Numbers / Tutoriaal Probleem Nommers

2017

11 Week: 15 May 2017 / 15 Mei 2017

All the indicated numbers refer to the “Problems” section in Serway; e.g. P1.9 is problem 9 from chapter 1.

Alle nommers verwys na die “Problems” afdeling in Serway; bv. P1.9 is probleem 9 uit hoofstuk 1.

P10.35 (a) $21.6 \text{ kg}\cdot\text{m}^2$ (b) $3.60 \text{ N}\cdot\text{m}$ (c) 52.5 revolutions

P10.36 (a) 1.95 s (b) If the pulley is massless, the acceleration would be larger by a factor $35/32.5$. The time is reduced by 3.6%.

P10.40 (a) Set $dI/dx = 0$ and solve for x and proof that this is the CoM (b) Proof

P10.45 (a) $143 \text{ kg}\cdot\text{m}^2$ (b) $2.57 \times 10^3 \text{ J}$

P10.49 $K_R = 1.04 \times 10^{-3} \text{ J}$

P10.50

$$v = \sqrt{\frac{2(m_1 - m_2)gh}{m_1 + m_2 + \frac{I}{R^2}}}$$
$$\omega = \frac{v}{R} = \sqrt{\frac{2(m_1 - m_2)gh}{m_1 R^2 + m_2 R^2 + I}}$$

P10.53 (a) $v = 1.59 \text{ m/s}$ (b) $\omega = 53.1 \text{ rad/s}$

P10.77 (a) Particle under a net force (b) Rigid object under a net torque (c)

$T_1 = 118 \text{ N}$ (d) $T_2 = 156 \text{ N}$ (e) symbolic (f) $1.17 \text{ kg}\cdot\text{m}^2$

P10.79

$$\omega = \sqrt{\frac{2mgd \sin \theta + kd^2}{I + mR^2}}$$